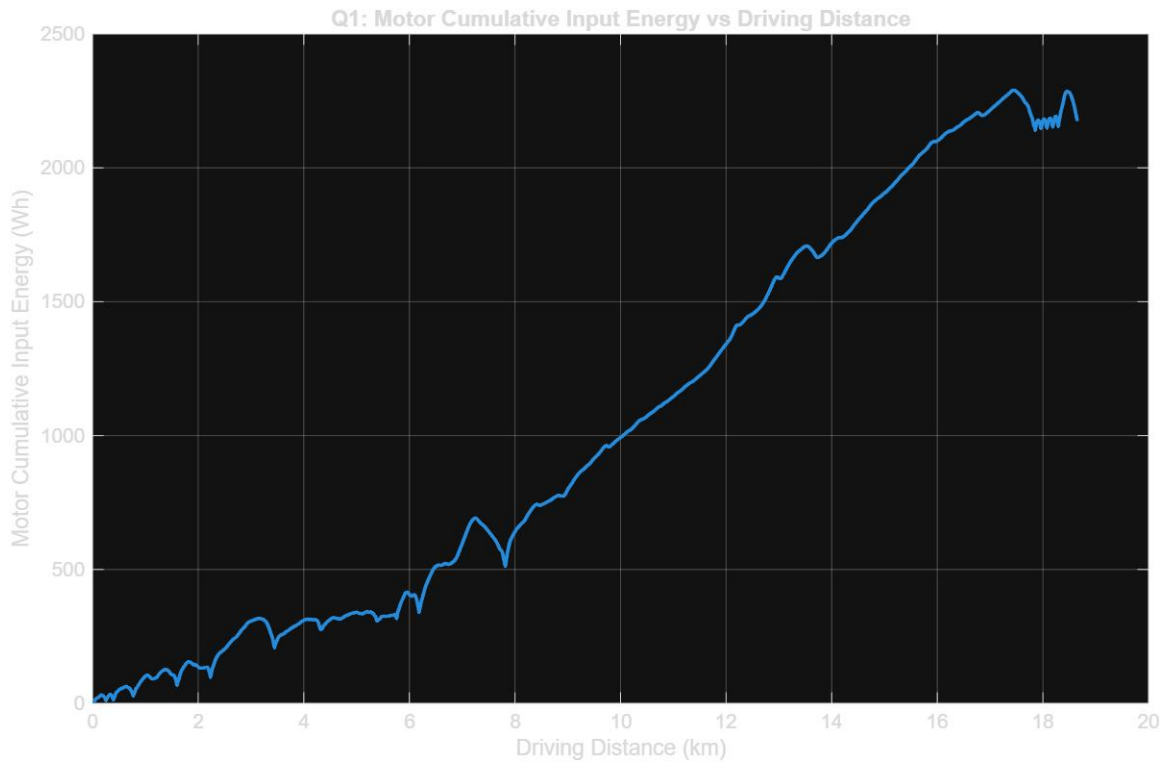


Q1:



This figure shows the motor's cumulative input energy consumption as the vehicle moves along the SC03+US06 drive cycle. The overall trend increases with driving distance, which is reasonable because energy is continuously consumed as the vehicle travels. The slope changes at different parts of the cycle, which means the vehicle does not consume energy at a constant rate. Small local fluctuations may appear because of speed changes, regenerative braking, or numerical effects, but the overall increasing trend makes sense.

CAL:

Total driving distance over the SC03 + US06 cycle:

$$\mathbf{D = 18.648 \text{ km}}$$

Motor input energy over the cycle:

$$\mathbf{E_{motor} = 2176.644 \text{ Wh}}$$

Ideal energy consumption:

$$\mathbf{\text{Energy consumption} = E_{motor} / D = 2176.644 / 18.648 = 116.721 \text{ Wh/km}}$$

Target vehicle range = 200 km

Required battery energy:

$$E_{\text{required}} = 116.721 \times 200 = 23344.2 \text{ Wh}$$

Battery cell parameters:

$$\text{Cell voltage} = 3.8 \text{ V} \quad \text{Cell capacity} = 2.2 \text{ Ah}$$

Energy per cell:

$$E_{\text{cell}} = 3.8 \times 2.2 = 8.36 \text{ Wh}$$

Series cells required for ~400 V bus:

$$N_s = 400 / 3.8 = 105.26 \approx 105$$

Total number of cells required:

$$N_{\text{total}} = 23344.2 / 8.36 = 2792.37$$

Parallel cells:

$$N_p = 2792.37 / 105 = 26.59 \approx 27$$

Final battery configuration:

$$N_s = 105 \quad N_p = 27$$

Total cells:

$$N_{\text{total}} = 105 \times 27 = 2835$$

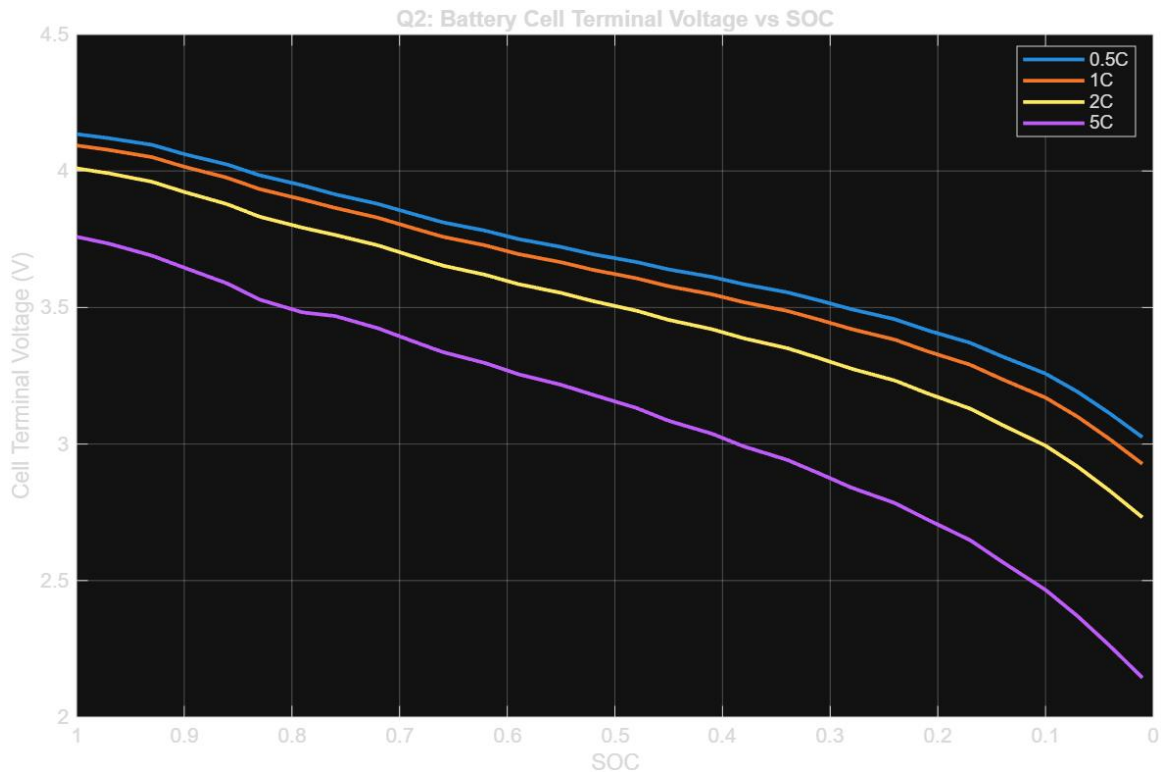
Nominal pack energy:

$$E_{\text{pack}} = 2835 \times 8.36 = 23700.6 \text{ Wh}$$

Battery pack capacity:

$$\text{Capacity} = 27 \times 2.2 = 59.4 \text{ Ah}$$

Q2:



CAL:

Battery terminal voltage model:

$$V_{\text{cell}} = V_{\text{oc}}(\text{SOC}) - I \times R(\text{SOC})$$

Cell capacity = 2.2 Ah

Current for each C-rate:

$$0.5C: I = 0.5 \times 2.2 = 1.1 \text{ A}$$

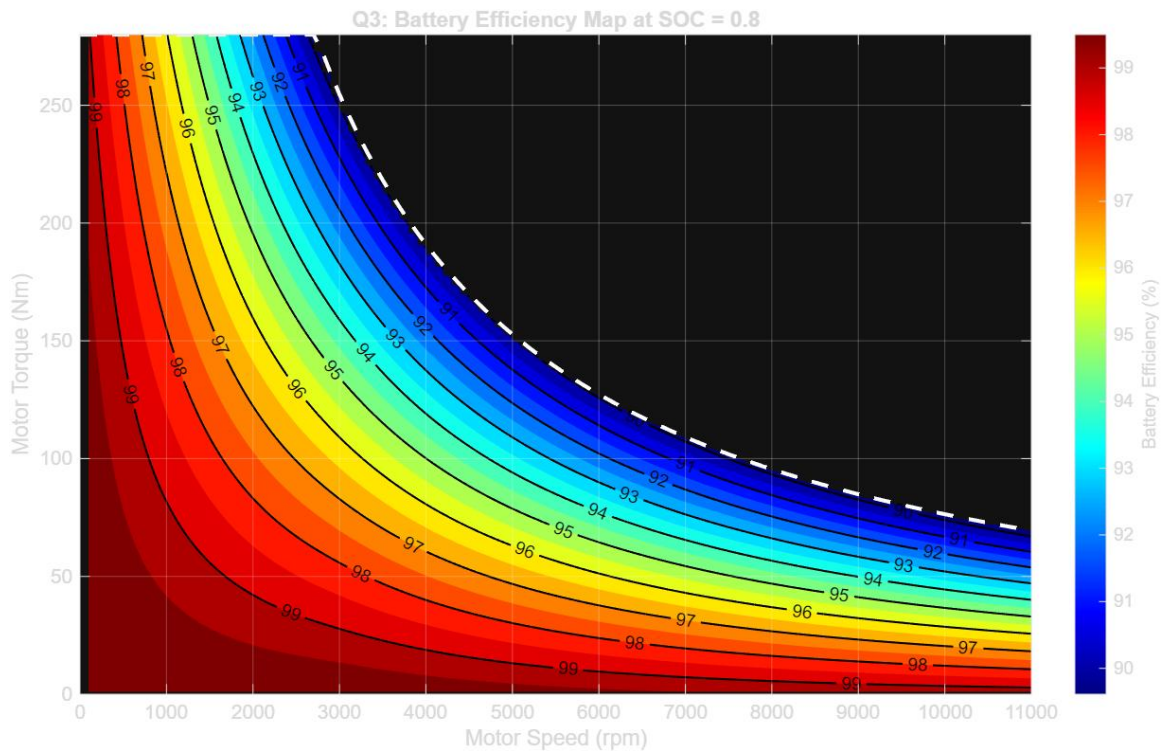
$$1C: I = 1 \times 2.2 = 2.2 \text{ A}$$

$$2C: I = 2 \times 2.2 = 4.4 \text{ A}$$

$$5C: I = 5 \times 2.2 = 11 \text{ A}$$

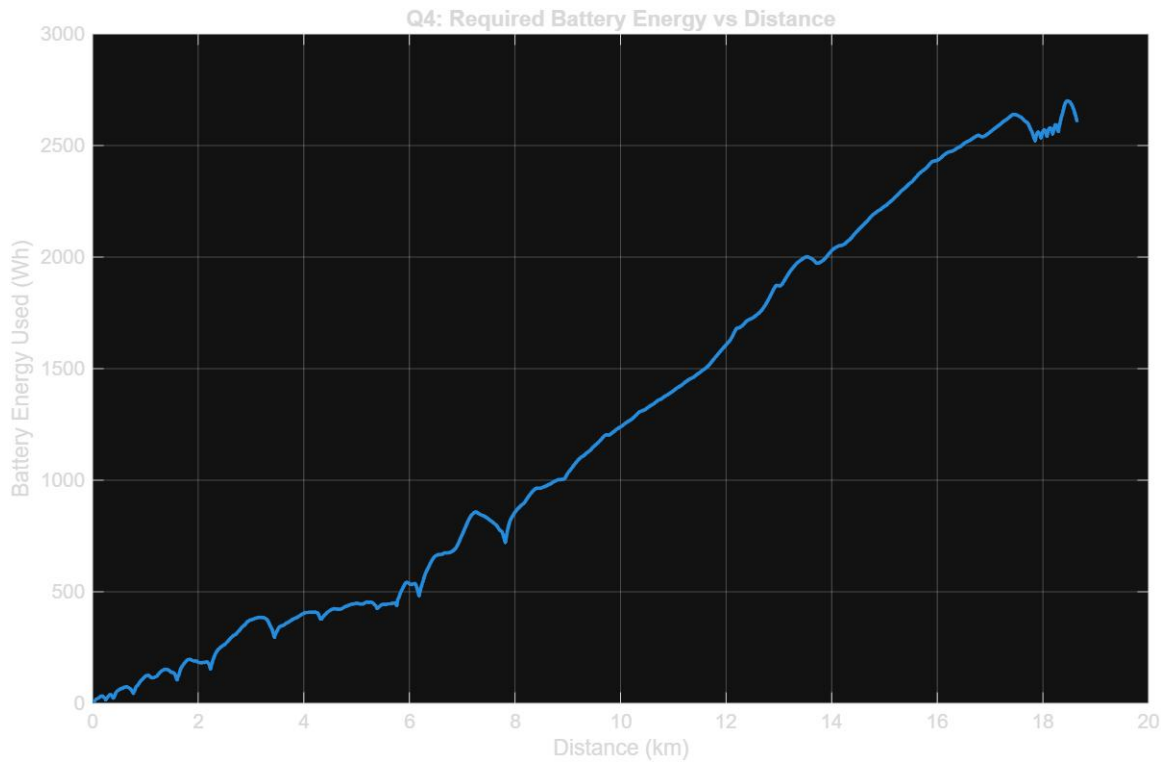
This figure shows how the battery cell terminal voltage changes with SOC under different discharge rates. As SOC decreases, the terminal voltage also decreases, which is expected for a lithium-ion battery. In addition, higher C-rates produce lower terminal voltage because a larger current causes a larger voltage drop across the internal resistance. The 5C curve is the lowest, while the 0.5C curve is the highest, so the result is physically reasonable and matches the expected battery behavior.

Q3:

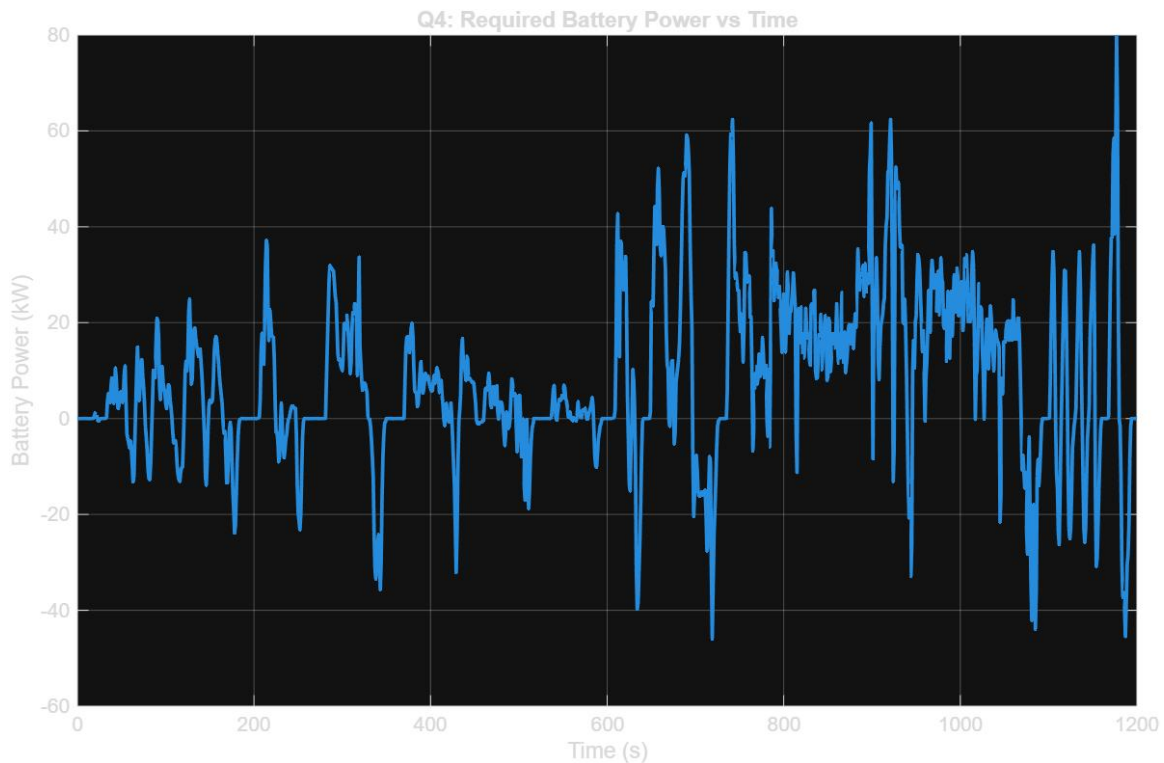


This figure presents the battery efficiency map at 0.8 SOC using motor speed as the x-axis and motor torque as the y-axis. The battery efficiency is higher in low-power operating regions and decreases as motor speed and torque increase. This trend makes sense because higher power demand causes larger battery current, and higher current produces larger resistive loss inside the battery. The contour labels and color variation clearly show the efficiency boundaries, and the overall shape is consistent with the expected effect of internal resistance on battery performance.

Q4:



This figure shows the required battery power over the SC03+US06 drive cycle. Positive battery power corresponds to battery discharge when the vehicle is driving or accelerating, while negative battery power indicates regenerative braking. The power changes significantly with time because the drive cycle includes many speed changes and transient conditions. This behavior is reasonable and shows that battery load is highly dynamic during real driving.



This figure shows the required battery energy as a function of driving distance when battery, motor, and power electronics losses are all included. The battery energy increases overall with distance, which is expected because the battery supplies energy to the system throughout the drive cycle. Compared with the motor-only energy in Question 1, this energy is higher because the model now includes battery internal loss and the 95% power electronics efficiency. Therefore, the updated vehicle energy consumption is higher, and the new estimated range becomes lower than the ideal range from Question 1.

CAL:

Actual battery energy used over drive cycle:

$$E_{\text{batt}} = 2607.467 \text{ Wh}$$

Drive distance:

$$D = 18.648 \text{ km}$$

Actual energy consumption:

$$\text{Wh/km} = 2607.467 / 18.648 = 139.824 \text{ Wh/km}$$

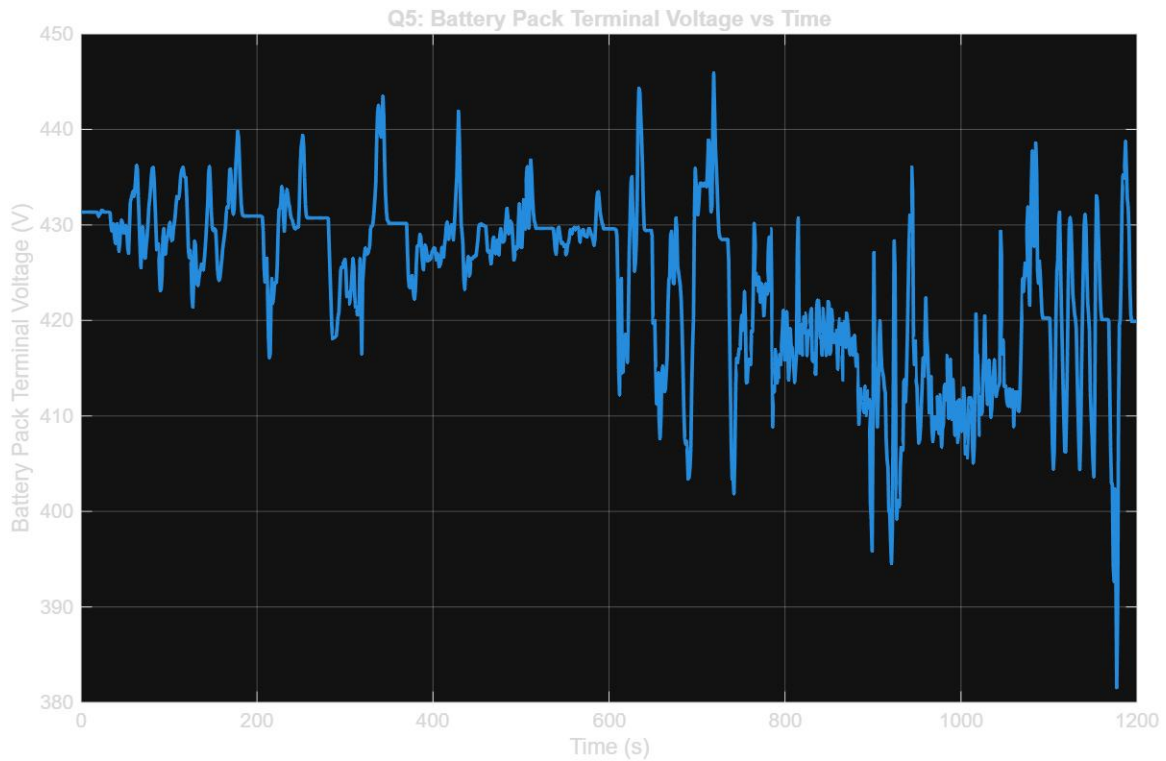
Battery pack energy from Q1:

$$E_{\text{pack}} = 2835 \times 8.36 = 23700.6 \text{ Wh}$$

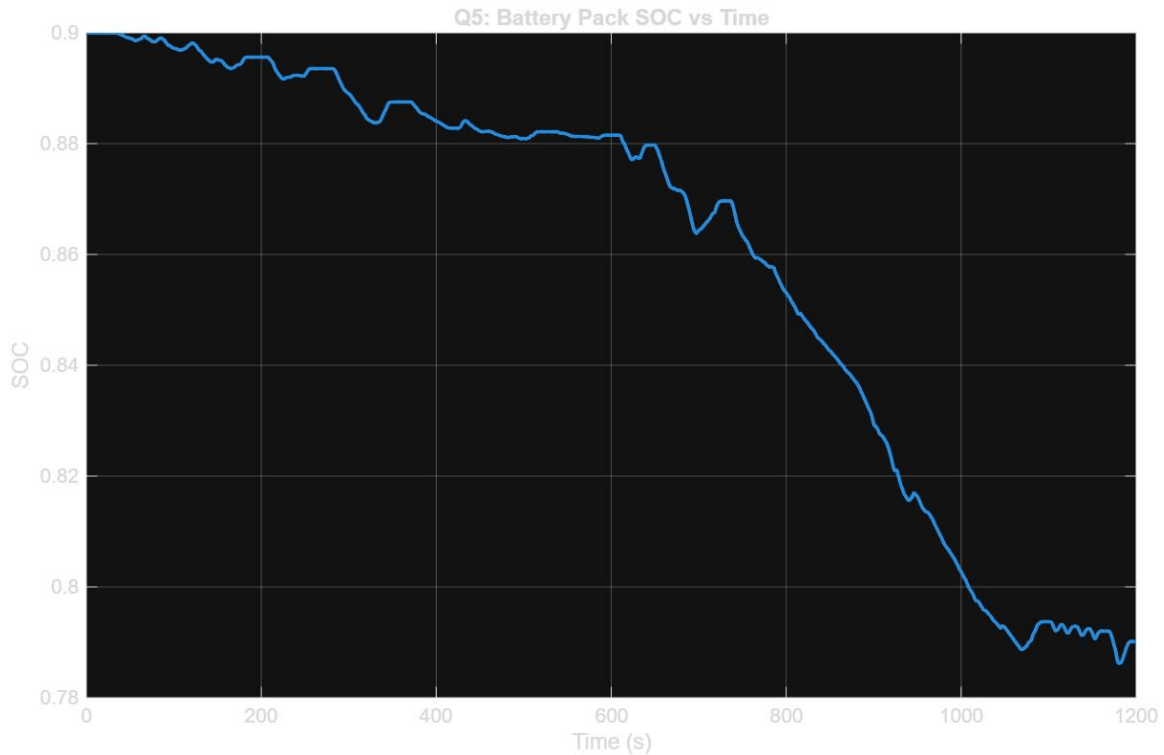
New vehicle range:

$$\text{Range} = 23700.6 / 139.824 = 169.503 \text{ km}$$

Q5:



This figure shows the variation of battery pack terminal voltage over the drive cycle. The terminal voltage fluctuates with time because it depends on both the battery SOC and the instantaneous current demand. During high-power operation, the current increases and the voltage drops because of the internal resistance term IR . During lighter load or regenerative events, the voltage becomes higher again, so the changing voltage trend in the figure is physically reasonable.



This figure shows how the battery pack SOC changes during one SC03+US06 combined drive cycle, starting from 0.9 SOC. The SOC decreases overall with time because the battery is supplying net energy to the vehicle. The slope is not constant, which means the battery is not discharging at a constant rate throughout the drive cycle. This result is reasonable because vehicle speed, torque demand, and battery power all change over time.

CAL:

Initial battery SOC:

$$\text{SOC}_{\text{initial}} = 0.9000$$

Final SOC after one SC03 + US06 cycle:

$$\text{SOC}_{\text{final}} = 0.7901$$

SOC drop:

$$\Delta \text{SOC} = 0.9000 - 0.7901 = 0.1099$$

Terminal voltage equation:

$$\mathbf{V_{pack} = V_{ocpack} - I \times R_{pack}}$$

As SOC decreases, open circuit voltage decreases and the pack voltage changes depending on load current.